v3: Environment Specification

1 Overview

The `v3` environment is a single-sector air traffic conflict resolution environment in which a reinforcement learning agent issues discrete heading instructions to maintain separation between en-route aircraft. Each episode places between ten and fifteen Airbus A320 aircraft in a randomly shaped convex sector at FL350. Aircraft enter from the sector boundary, transit toward a destination beyond the opposite boundary, and are replaced when they exit, keeping traffic density approximately constant. The agent observes the state of the focus aircraft and its four most urgent neighbours and selects one of eight discrete actions per step.

The action space is structured around the ATC resolution pattern of turn, hold, and return. Six *turn instructions* deflect the focus aircraft's commanded heading by $\delta \in \{\pm 30^\circ, \pm 45^\circ, \pm 60^\circ\}$ from the *current commanded heading*, and are stackable: issuing two consecutive left-30° turns accumulates a total -60° deviation. A *hold* action maintains the current commanded heading without additional cost, allowing the aircraft to execute a turn over multiple steps. A *fly-direct* action returns the commanded heading to the destination bearing, eliminating accumulated drift once the conflict has resolved.

The reward signal is purely negative: a large penalty is applied when a separation violation occurs, a graduated conflict penalty scales with the imminence and predicted miss distance of approaching pairs, a drift penalty accumulates while the aircraft is held off its planned route, and a workload penalty is charged once per turn instruction. The combination of drift and workload terms creates a sustained incentive to resolve conflicts with as few instructions and as little deviation as possible.

The focus aircraft, the one to which the instruction is applied, is selected each step by an urgency-based mechanism that prioritises the aircraft involved in the most severe pair conflict and locks onto a single aircraft for the duration of a resolution sequence. When the sector is conflict-free, focus shifts to route recovery: the most deviated aircraft with sufficient surrounding airspace is selected, so that it can be returned to its route without creating a new conflict.

2 Scenario

Episodes are simulated using the BlueSky air traffic simulator at a physics timestep of $\Delta t = 0.5\,\mathrm{s}$, with a flat-earth equatorial projection (reference latitude 0°, so the longitude scale factor equals unity everywhere). At the start of each episode, $N \sim \mathcal{U}\{10, \ldots, 15\}$ Airbus A320 aircraft are initialised at FL350 and cruise at a fixed speed of $M = 0.78$ ($\approx 450\,\mathrm{kt}$ TAS) throughout. The sector boundary is a random convex polygon with five to seven vertices, scaled to an area of $N/\rho\,\mathrm{km}^2$, where $\rho \sim \mathcal{U}(1/15\,000,\ 1/5\,000)\,\mathrm{aircraft/km}^2$ (equivalent to $5\,000$–$15\,000\,\mathrm{km}^2$ per aircraft). Polygons with an isoperimetric ratio below 0.65 are rejected to avoid pathologically elongated sectors. Each aircraft is spawned at a distinct arc of the sector perimeter with uniform jitter, assigned a heading toward a reference point on the opposite arc, and given a destination two sector diameters beyond the boundary. The reference point marks the intended boundary exit of the aircraft and is retained for the route-adherence metric described in Section ??. Candidate spawns are screened by the urgency measure of Equation ??: a placement is rejected when it would create a pair urgency above $u_{\text{spawn}} = 0.50$ with any active aircraft. Since urgency exceeds one inside the separation standard, this single rule excludes both spawns in violation of separation and spawns on a collision course with a predicted conflict closer than 180 s, so no aircraft enters the sector in an immediate conflict. A rejected placement is retried at a different boundary position, and a slot for which no admissible placement is found is requeued for a later attempt. Aircraft that exit the sector are replaced within one step, keeping traffic density approximately constant throughout each episode. Episode length corresponds to approximately 60 minutes of simulated time, derived from the sector diameter and nominal cruise speed.

Each RL step advances the simulation by $10 \times 0.5\,\text{s} = 5\,\text{s}$. The selected action is applied at the start of the step and one instruction is issued to the *focus aircraft*, selected as described in Section ??. Separation is checked at the end of the final sub-step.

3 Focus Selection

Because only one instruction is issued per step, a principled selection mechanism is required to determine which aircraft receives it. An urgency matrix $\mathbf{U} \in \mathbb{R}^{N \times N}$ is recomputed after every step from current positions and velocities. The entry u_{ij} is defined as

$$u_{ij} = \begin{cases} 1 + 9\left(1 - \dfrac{d_{ij}}{d_{\text{sep}}}\right) & d_{ij} < d_{\text{sep}} \\ \dfrac{T_{\text{warn}} - t_{\text{CPA}}^{ij}}{T_{\text{warn}}} & d_{\text{CPA}}^{ij} < d_{\text{sep}}, \quad 0 \le t_{\text{CPA}}^{ij} \le T_{\text{look}} \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

with $d_{\text{sep}} = 5\,\text{NM}$, $T_{\text{warn}} = 360\,\text{s}$, and $T_{\text{look}} = 900\,\text{s}$. The active-violation branch scales linearly from $u_{ij} = 1$ at the separation boundary to $u_{ij} = 10$ at zero distance. The predicted-conflict branch ramps linearly from 0 at T_{warn} to 1 at $t_{\text{CPA}} = 0$; both branches are continuous at $d_{ij} = d_{\text{sep}}$.

The focus aircraft is selected as the aircraft involved in the most severe pair conflict:

$$f = \arg\max_i \max_{j \neq i} u_{ij}, \tag{2}$$

with ties broken by the total urgency burden $\sum_{j \neq i} u_{ij}$. The tiebreak matters when both members of the worst conflict pair share the same pair urgency: the aircraft additionally burdened by other conflicts is then preferred, an important distinction at high traffic densities.

Route recovery. When no conflicts are active, the selection falls back to a route-recovery mode. This mode is necessary because instructions are only ever issued to the focus aircraft. An aircraft that has been turned away from its route during conflict resolution keeps flying its commanded heading indefinitely; unless it regains focus at a later step, it is never instructed to return and drifts progressively further from its intended track. The recovery mode therefore uses conflict-free periods to direct focus to the aircraft most in need of being returned.

Each candidate i is scored as the product of a drift factor and a clearance factor,

$$s_i = \underbrace{\frac{1 - \cos\left(\psi_{\text{dest}}^{(i)} - \psi_{\text{cmd}}^{(i)}\right)}{2}}_{\text{drift}} \times \underbrace{\min\left(1,\ \frac{d_i^{\text{nn}}}{d_{\text{clear}}}\right)}_{\text{clearance}}, \tag{3}$$

where d_i^{nn} is the distance to the nearest other aircraft and $d_{\mathrm{clear}} = 20\,\mathrm{NM}$, four times the separation standard. At the worst-case head-on closure rate of approximately 900 kt, this distance corresponds to 80 s to a potential separation violation, well beyond the roughly 20 s required to execute a 60° return turn. The drift factor is identical in form to the drift penalty of the reward function (Section **??**). Since that penalty is charged only for the focus aircraft, this choice directs focus to the aircraft whose deviation generates the largest per-step penalty, which gives the agent the strongest incentive to issue the fly-direct correction. The drift factor grows quadratically for small deviations, reaching 0.07 at 30° and 0.25 at 60°.

The clearance factor accounts for the fact that a return manoeuvre is itself a trajectory change and can create a new conflict. An aircraft whose nearest neighbour lies beyond d_{clear} receives full priority, while the priority of a deviated aircraft that is still boxed in by surrounding traffic is scaled down linearly, deferring its recovery until the local traffic has thinned out.

The aircraft with the highest score receives focus, subject to a hysteresis margin of 0.01: the current focus is retained unless another aircraft exceeds its score by this margin, which prevents oscillation between similarly scored candidates. The margin is small relative to the score scale, a 30° deviation scores 0.067, so that mildly deviated aircraft can still take focus from an aircraft whose recovery has completed. Once selected, the commitment mechanism described below holds focus for at least n_{clear} steps, giving the agent time to issue the correction and observe its effect before focus moves on.

Commitment. Conflict resolution typically requires several consecutive instructions to the same aircraft. The focus is therefore locked on f while $\max_{j \neq f} u_{fj} > 0$ or fewer than $n_{\mathrm{clear}} = 5$ steps have elapsed since the urgency last cleared. An emergency override releases the lock immediately when any other aircraft reaches $\max_j u_{ij} \geq u_{\mathrm{emerg}} = 0.8$, corresponding to approximately 72 s to predicted conflict.

4 Action Space

The action space is discrete with eight elements, summarised in Table **??**. The eight actions fall into three groups.

Turn instructions (actions 0–2 and 4–6). Six actions increment the commanded heading of the focus aircraft relative to the *current commanded heading*:

$$\psi_{\mathrm{cmd}} \leftarrow \big(\psi_{\mathrm{cmd}} + \delta\big) \bmod 360^\circ, \qquad \delta \in \{-60^\circ, -45^\circ, -30^\circ, +30^\circ, +45^\circ, +60^\circ\}. \tag{4}$$

Unlike a one-shot offset from the destination bearing, consecutive turn instructions are *stackable*: issuing the same turn twice accumulates the full 2δ deviation. This gives the agent finer-grained control over large avoidance manoeuvres and avoids the implicit reset to ψ_{dest} that would occur if the reference were recomputed at each step. A workload cost c is charged once when the instruction is issued; larger deviations carry higher cost.

Hold (action 3). Hold is a true no-operation: no instruction is issued and nothing is sent to the simulator. The aircraft simply continues executing its most recent instruction, which allows a previously commanded turn to complete over multiple steps without additional cost. It is the only action with zero cost and serves as the natural *no-intervention* baseline: if hold is applied every step, the controller stays silent and all aircraft follow their last commanded headings unmodified.

Fly direct (action 7). Fly direct resets $\psi_{\text{cmd}} \leftarrow \psi_{\text{dest}}(f)$, returning the aircraft to its planned route in a single step. The cost scales with the commanded deviation being corrected, so that fly direct is effectively free when the aircraft is already on track and expensive when a large turn is being aborted prematurely.

The three groups encode the intended ATC resolution pattern: issue one or more stackable turns to build up lateral separation, hold while the aircraft manoeuvres and the conflict geometry resolves, then fly direct to eliminate accumulated route deviation.

Table 1: Action space and associated workload costs.

Index	**Action**	**Cost** c
0	$\delta = -60°$	1.0
1	$\delta = -45°$	0.75
2	$\delta = -30°$	0.5
3	Hold (no instruction issued)	0.0
4	$\delta = +30°$	0.5
5	$\delta = +45°$	0.75
6	$\delta = +60°$	1.0
7	Fly direct ($\psi_{\text{cmd}} \leftarrow \psi_{\text{dest}}$)	$\lvert\Delta\psi_{\text{cmd}}\rvert/60° \in [0, 1]$

5 Observation Space

The observation vector $\mathbf{o} \in \mathbb{R}^{23}$ is constructed ego-centrically from the focus aircraft f. It comprises three ownship features and four intruder slots of five features each. All positional quantities are expressed in f's heading frame, with the forward axis aligned with ψ_{act} and the lateral axis 90° to the right. Intruder positions are encoded in *local polar coordinates* rather than Cartesian components, which provides rotational equivariance and avoids the separate normalisation of east and north offsets.

Table 2: Observation vector ($\mathbf{o} \in \mathbb{R}^{23}$). $d_{\text{warn}} = T_{\text{warn}}\, v_{\text{nom}} = 45\,\text{NM}$. CPA distance is normalised by d_{sep}, the correct scale for conflict geometry.

Index	**Symbol**	**Definition**	**Range**
Ownship state			
0	$\sin\Delta\psi$	$\sin(\psi_{\text{dest}} - \psi_{\text{cmd}})$	$[-1, 1]$
1	$\cos\Delta\psi$	$\cos(\psi_{\text{dest}} - \psi_{\text{cmd}})$	$[-1, 1]$
2	p_{turn}	$\text{wrap}(\psi_{\text{cmd}} - \psi_{\text{act}})/180°$	$[-1, 1]$
Per intruder j (slots 1–4, offset $+k$ for slot k)			
+0	r_{fj}	d_{fj}/d_{warn}	$[0, 1]$
+1	θ_{fj}	$\text{atan2}(\Delta e_{fj}^{\text{ego}},\, \Delta n_{fj}^{\text{ego}})/\pi$	$[-1, 1]$
+2	r_{fj}^{CPA}	$d_{fj}^{\text{CPA}}/d_{\text{sep}}$	$[0, 1]$
+3	θ_{fj}^{CPA}	$\text{atan2}(\Delta e_{fj}^{\text{ego,CPA}},\, \Delta n_{fj}^{\text{ego,CPA}})/\pi$	$[-1, 1]$
+4	$\hat{t}_{\text{CPA}}$	$\text{clip}(t_{\text{CPA}}/T_{\text{warn}},\, 0,\, 1)$	$[0, 1]$

The heading error is encoded as the pair $(\sin\Delta\psi,\, \cos\Delta\psi)$ rather than a scalar angle, which avoids the discontinuity at ±180°. The error is measured against the *commanded* heading ψ_{cmd} rather than the actual heading, consistent with the drift penalty in the reward function. Turn progress p_{turn} captures the execution lag: it is zero when the aircraft has completed its commanded turn and non-zero while the physical heading still lags behind the command. The difference is wrapped to $(-180°, 180°]$ and normalised by $180°$, so the feature is bounded without clipping even when several stacked turns produce a large commanded deviation.

For each intruder j, the relative position is encoded in polar form. The range $r_{fj} \in [0, 1]$ gives the normalised distance. The bearing $\theta_{fj} \in [-1, 1]$ is zero when the intruder is directly ahead, $+1$ (or -1) when it is directly behind, and ± 0.5 when it is abeam; it is defined as $\operatorname{atan2}(\Delta e^{\text{ego}}_{fj}, \Delta n^{\text{ego}}_{fj})/\pi$, where Δe^{ego} and Δn^{ego} are the lateral (right) and forward components of the relative displacement in f's heading frame. This polar encoding preserves the separation of range and bearing information and makes the sign of θ_{fj} a direct indicator of which side the intruder lies on.

The CPA block uses the same polar convention. The miss-distance $r^{\text{CPA}}_{fj} \in [0, 1]$ is the predicted closest approach normalised by d_{sep}; it is zero for non-converging pairs. The bearing θ^{CPA}_{fj} identifies which side the intruder will pass: a negative value indicates a left-side predicted conflict, providing a direct signal for turning direction. The normalised time $\hat{t}_{\text{CPA}}$ mirrors the imminence factor in the reward function. A value of 1 signals that no conflict is predicted, making the CPA features uninformative (both zero). Non-converging pairs and empty slots receive the sentinel (1, 0, 0, 0, 1).

The CPA displacement in the heading frame is

$$\begin{pmatrix} \Delta e^{\text{ego,CPA}}_{fj} \\ \Delta n^{\text{ego,CPA}}_{fj} \end{pmatrix} = R(\psi_{\text{act}}) \left[\begin{pmatrix} \Delta e_{fj} \\ \Delta n_{fj} \end{pmatrix} + t_{\text{CPA}} \begin{pmatrix} \Delta v_{e,fj} \\ \Delta v_{n,fj} \end{pmatrix} \right], \tag{5}$$

where $R(\psi)$ rotates geographic (east, north) coordinates into the ego-centric (lateral, forward) heading frame.

5.1 Intruder Slot Ordering

The four intruder slots are filled in two passes. In the first pass, all aircraft j for which $u_{fj} > 0$ are inserted in descending urgency order until the slots are full. In the second pass, any remaining slots are filled by the nearest aircraft by current distance, excluding those already selected. This two-pass strategy ensures that predicted-conflict aircraft are prioritised while nearby diverging aircraft, which may become relevant within the current step, are guaranteed a slot when space permits.

6 Reward Function

The reward is purely negative at every step:

$$r_t \;=\; r_{\text{LoS}} + r_{\text{conflict}} + r_{\text{drift}} + r_{\text{work}} \;\leq\; 0. \tag{6}$$

By excluding positive terms, the training objective reduces to minimising cumulative harm, and the optimal policy is the one that keeps all penalties as small as possible throughout the episode. The four components and their weights are summarised in Table **??**.

Loss-of-separation penalty. A separation violation is detected when any pair of aircraft is found within d_{sep} of each other at the end of the 5-second simulation window:

$$r_{\text{LoS}} = -w_{\text{LoS}} \cdot \mathbf{1}[\ell_{\text{LoS}}]. \tag{7}$$

This component carries the largest weight and signals that a violation has already occurred, as distinct from the graduated conflict signal below. The check is sector-wide: violations between any pair, not only those involving the focus aircraft, contribute to ℓ_{LoS}.

Conflict penalty. For each converging intruder k whose predicted closest-point-of-approach satisfies $d_{\text{CPA}}^{(k)} < d_{\text{sep}}$, a severity score is computed as the product of an imminence factor and a miss-distance factor:

$$s_k \;=\; \max\left(0,\; 1 - \frac{t_{\text{CPA}}^{(k)}}{T_{\text{warn}}}\right) \times \max\left(0,\; 1 - \frac{d_{\text{CPA}}^{(k)}}{d_{\text{sep}}}\right), \tag{8}$$

and the penalty is taken over the worst-case intruder:

$$r_{\text{conflict}} = -w_{\text{conflict}} \cdot \max_k \, s_k. \tag{9}$$

The imminence factor is zero when $t_{\text{CPA}} \geq T_{\text{warn}}$ and rises to 1 as the conflict becomes imminent. The miss-distance factor is zero when $d_{\text{CPA}} = d_{\text{sep}}$ and rises to 1 as the predicted miss distance approaches zero, so that a grazing predicted conflict scores lower than a direct collision course at the same time horizon. The predicted-LoS gate ($d_{\text{CPA}} < d_{\text{sep}}$) ensures that only pairs on a true collision course contribute. Aircraft already in active separation violation contribute $s_k = 1$.

Drift penalty. A continuous penalty on the commanded-heading deviation from the destination bearing penalises the agent for holding the aircraft off its planned route:

$$r_{\text{drift}} = -w_{\text{drift}} \cdot \frac{1 - \cos(\psi_{\text{dest}} - \psi_{\text{cmd}})}{2}. \tag{10}$$

Unlike the workload term, drift is charged every step that the commanded heading deviates from the destination bearing, regardless of which action was issued. During the hold phase this penalty accumulates, creating a sustained incentive to issue the fly-direct action once a conflict has been resolved.

Workload penalty. A one-time cost is charged each time a turn instruction or a fly-direct instruction is issued. Turn costs are proportional to the heading change:

$$c = \frac{|\delta|}{60^\circ}, \qquad r_{\text{work}} = -w_{\text{work}} \cdot c. \tag{11}$$

This ensures that the total workload cost depends only on the accumulated heading excursion and not on how many instructions were used to achieve it: two consecutive -30° turns ($2 \times 0.5 = 1.0$) cost the same as one -60° turn (1.0). The hold action (action 3) is the only truly free action ($c = 0$).

For the fly-direct action (action 7) the cost scales with the commanded deviation being corrected at the moment the instruction is issued:

$$r_{\text{work}}^{\text{direct}} = -w_{\text{work}} \cdot \min\left(\frac{|\Delta\psi_{\text{cmd}}|}{60^\circ},\, 1\right), \qquad \Delta\psi_{\text{cmd}} = \psi_{\text{dest}} - \psi_{\text{cmd}}^{-}, \tag{12}$$

where ψ_{cmd}^{-} denotes the commanded heading immediately before the fly-direct is applied. The cost is capped at 1 to handle deviations larger than 60° that can accumulate from multiple consecutive turns.

Table 3: Reward weights in descending order of magnitude.

Weight	Penalises	Value
w_{LoS}	Separation violation at step end	10.00
w_{conflict}	Worst-case imminence $\times$ miss-distance severity ($d_{\text{CPA}} < d_{\text{sep}}$)	3.00
w_{drift}	Per-step deviation; accumulates during hold, motivates fly-direct	1.00
w_{work}	Per-turn cost $= w_{\text{work}} \cdot \lvert\delta\rvert/60°$; fly-direct cost $= w_{\text{work}} \cdot \min(\lvert\Delta\psi\rvert/60°, 1)$; hold free	1.00

7 Training Configuration

The policy is trained with Proximal Policy Optimisation (PPO) as implemented in Stable-Baselines3, using four parallel environments (`SubprocVecEnv`) each seeded independently. Observation normalisation is disabled (`norm_obs=False`) because all features are bounded by design. Reward normalisation is enabled (`norm_reward=True`): rewards are divided by the running standard deviation of the discounted return, which stabilises value function learning in the presence of sparse, large loss-of-separation penalties. The policy and value networks share a two-layer MLP with [128, 128] hidden units.

Table 4: PPO hyperparameters.

Symbol	Parameter	Value
γ	Discount factor	0.99
λ	GAE parameter	0.95
T	Rollout steps per environment	1 024
N_{envs}	Parallel environments	4
B	Mini-batch size	64
K	Epochs per update	10
ϵ_{clip}	PPO clipping range	0.2
β_H	Entropy coefficient	0.02
α	Learning rate	3×10^{-4}
	Network architecture	[128, 128]

8 Validation

Policy performance is evaluated by sweeping a two-dimensional grid of operating conditions: aircraft density ρ on one axis and traffic count N on the other. The sweep parameters are summarised in Table **??**. These ranges cover the high-density, high-count conditions that are most relevant to operational stress testing, and extend beyond the training distribution ($N \sim \mathcal{U}\{10, \ldots, 15\}$) to assess out-of-distribution robustness at both ends.

Table 5: Validation sweep parameters. Density values are given in aircraft/km^2; the parenthesised values show the equivalent area per aircraft in km^2.

Parameter	Values			
Density ρ (aircraft/km^2)	$2{\times}10^{-4}$ (5 k)	$1.25{\times}10^{-4}$ (8 k)	10^{-4} (10 k)	$6.7{\times}10^{-5}$ (15 k)
Traffic count N (aircraft)	8 10 12 14			
Episodes per cell	8			
Checkpoint	4.6 M training steps			

The primary metric is the *loss-of-separation fraction*: the fraction of steps within an episode in which at least one separation violation ($d < d_{\mathrm{sep}}$) is active. Averaging over eight episodes per cell reduces variance from random sector geometry and initial placement.

Route adherence is tracked as a secondary metric, the *arrival rate*. An exit is counted as on-target when the aircraft leaves the sector within 5 NM of the reference exit point assigned at spawn, and the arrival rate is the fraction of on-target exits per episode. Under the no-intervention baseline the arrival rate equals one by construction, since undisturbed aircraft fly straight to their reference point. A reduced arrival rate therefore isolates the route inefficiency introduced by conflict-resolution manoeuvres that are not followed by a timely return to track. Both metrics are also logged per episode during training.

The same grid is evaluated under a *no-policy baseline* in which actions are drawn uniformly at random. This baseline measures the separation violation rate that would occur without any conflict resolution, so that the reduction attributable to the learned policy can be quantified directly.

Results are presented as side-by-side heatmaps of mean loss-of-separation fraction for the policy and the baseline, using a shared colour scale to allow direct visual comparison. A second figure shows loss-of-separation fraction as a function of density, with one panel per traffic count and shaded bands indicating ± 1 standard deviation across episodes.